

ASWIN BABU K M

Database Developer | *Linux Systems* | *DevOps Engineer*

Madurai, India | +91 9342489851 | ashwinbabu769@gmail.com | [linkedin.com/in/aswin-babu-3baab235a](https://www.linkedin.com/in/aswin-babu-3baab235a) | <https://github.com/Aswinbabu5> | <https://portfolio-nlb1.vercel.app>

PROFESSIONAL SUMMARY

DevOps Engineer with hands-on experience in Linux production support, infrastructure monitoring, and troubleshooting distributed systems at Zoho. Built internal monitoring proof-of-concepts, investigated production incidents through log analysis, and worked with Linux internals, eBPF, networking, and automation. Strong backend development knowledge in Java, Python, Spring Boot, and REST API development.

TECHNICAL SKILLS

Databases: SQL, Stored Procedure, Functions, Triggers, Indexing, Joins, Query Optimisation.

Monitoring: System Monitoring, Log Analysis, Grafana, Prometheus.

DevOps & Systems: Linux Administration (RHEL/Ubuntu), Shell Scripting, Docker, Kubernetes, CI/CD Concepts.

Networking: TCP/IP, Packet Processing, eBPF.

Programming language: Python, Java, C/C++, Bash.

AI/GEN-AI: LLM, Semantic search, vectorDB, Embedding.

Backend & APIs: Spring Boot, REST APIs, Struts.

Frontend: HTML, CSS, JavaScript.

Tools & Platforms: Git, Postman, VSCode.

Concepts: linux, Data Structures, OOP.

PROFESSIONAL EXPERIENCE

ZOHO CORPORATION PVT LTD — *Chengalpattu, India Technical Trainee*

September 2025 – March 2026

- Monitored production infrastructure across multiple data centers, investigated production incidents through log analysis, and collaborated with engineering teams to reduce service downtime.
- Investigated and resolved application, networking, and infrastructure-level production incidents through systematic log analysis and root cause identification.
- Collaborated with cross-functional engineering teams to diagnose complex issues, reduce mean time to resolution, and implement preventive fixes.
- Built a proof-of-concept monitoring solution for tracking feature flag deployment status across distributed data centers, enabling faster operational visibility and reducing deployment rollback time.

Internship Trainee — Linux Systems

July 2025 – August 2025

- Gained hands-on experience with Linux system calls, including deep-dive experiments with `io_uring` for asynchronous I/O optimization.
- Explored eBPF (Extended Berkeley Packet Filter) for packet processing and network interface operations, enabling performance monitoring and low-level network troubleshooting.
- Conducted User-level experiments involving context switching, system call execution, and strengthening system-level debugging capabilities.

EDUCATION

B.Tech in Information Technology

K.L.N College of Engineering, Sivagangai

November 2022 – April 2026

Higher Secondary Certificate (HSC)

Sourashtra Boys Higher Secondary School, Madurai

September 2021 – May 2022

Secondary School Leaving Certificate (SSLC)

Seventh-Day Adventist Matriculation Hr Sec School, Madurai

June 2019 – April 2020

PROJECTS

The Judicial Assistant — Jubilee

July 2025 – December 2025

- Developed an ML-powered automation pipeline in Python to assist users in retrieving and analyzing legal case records, demonstrating data processing and backend scripting capabilities.
- Applied NLP techniques for intelligent search and filtering of structured legal datasets, improving information retrieval efficiency.
- Implemented a voice inputs, translators, read Aloud, toolkit to describe or speech the output.
- **Technologies:** Python, NLP, Scikit-learn, Pandas, Machine Learning
- <https://jublie.streamlit.app>

Smart Air Quality Monitoring for Asthma Patients

January 2026 – April 2026

- Built an IoT-based air quality monitoring system using Raspberry Pi and ESP32.
- Integrated four environmental sensors.
- Implemented Wi-Fi communication for real-time data transfer.
- Developed Python scripts for data collection and monitoring.
- Achieved real-time transmission with latency under 500 ms.
- **Technologies:** Python, Flutter, Raspberry pi 4, ESP 32, C++

Developer Dashboard

July 2026 – August 2026

- Developed a responsive developer dashboard with project tracking, Kanban board, analytics, and team management.
- Built reusable React components and responsive user interfaces.
- Applied routing, component-based architecture, and state management principles.
- **Technologies:** React.js, LocalStorage, BarCharts, Lucide-react
- <https://developer-dashboard-tan.vercel.app/>

CERTIFICATIONS

- Backend Developer — Meta (Coursera)
- Bug Bounty — Prompt Infotech
- Python Programming — SSI Institution
- Power BI and Tableau — Syasans
- C Programming Training — IIT Spoken Tutorial
- Smart India Hackathon (PRE-SIH 2024) — Participant